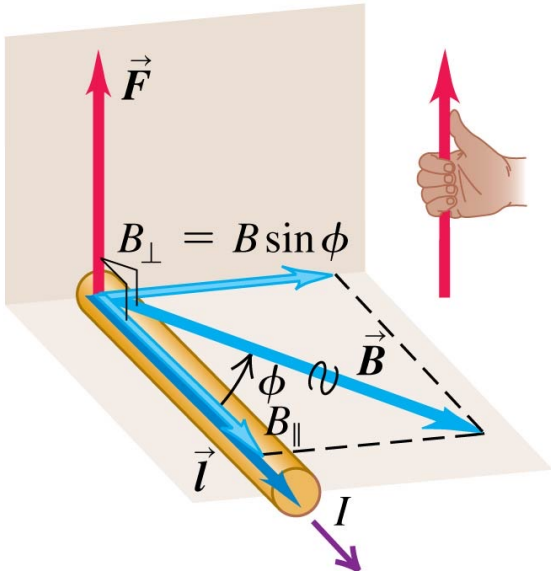
Magnetic force on a current-carrying conductor

- The force is always perpendicular to both the conductor and the field, with the direction determined by the same **right-hand rule** used for a moving positive charge.

Force $\vec{F}$ on a straight wire carrying a positive current and oriented at an angle ϕ to a magnetic field $\vec{B}$:

- Magnitude is $F = IlB_{\perp} = IlB\sin\phi$.
- Direction of $\vec{F}$ is given by the right-hand rule.



Magnetic force on a straight wire segment

$$\vec{F} = I\vec{l} \times \vec{B}$$

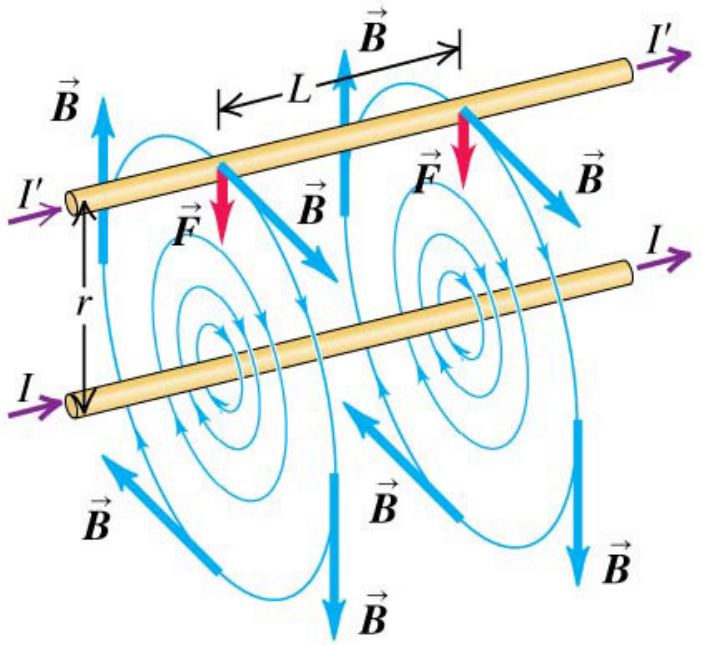
Current

Magnetic field

Vector length of segment (points in current direction)

Force between parallel conductors

- Two long, straight, parallel conductors separated by a distance r and carrying currents I and I' in the same direction
- Each conductor lies in the magnetic field set up by the other, so each experiences a force.



$$B = \frac{\mu_0 I}{2\pi r} \qquad \vec{F} = I'\vec{L} \times \vec{B} \qquad F = I'LB = \frac{\mu_0 I I' L}{2\pi r}$$

Magnetic force per unit length between two long, parallel, current-carrying conductors

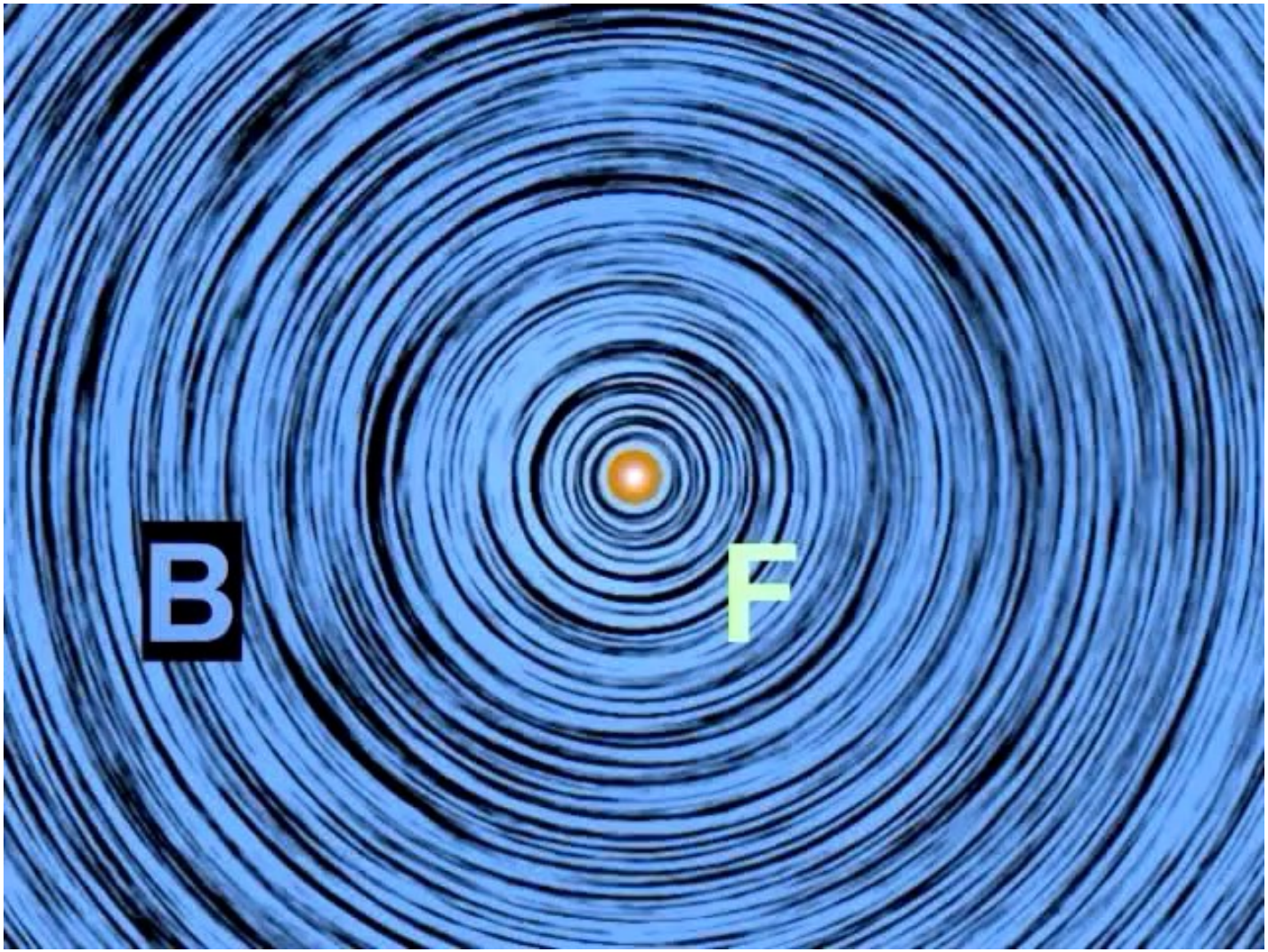
$$\frac{F}{L} = \frac{\mu_0 I I'}{2\pi r}$$

- μ_0: Magnetic constant
- I: Current in first conductor
- I': Current in second conductor
- r: Distance between conductors

© 2016 Pearson Education, Ltd.

Question 12

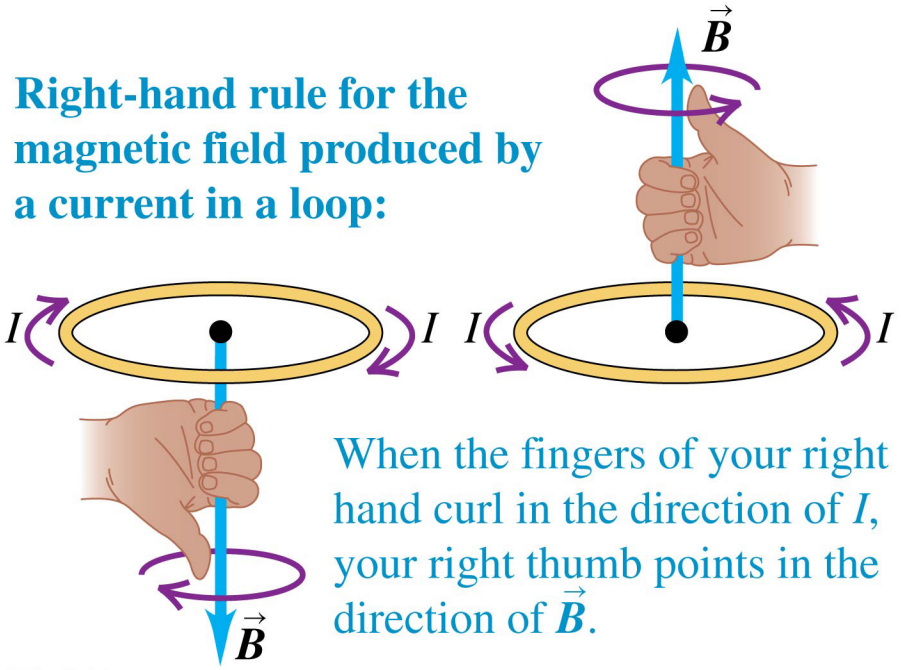
- What is the current direction?

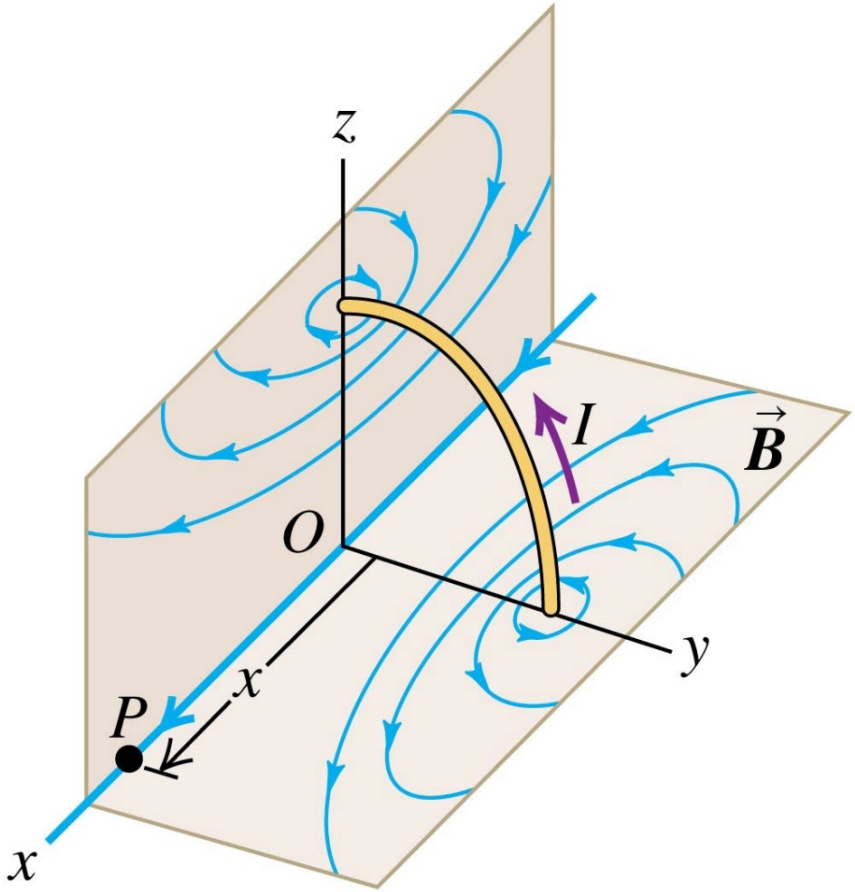


© 2016 Pearson Education, Ltd.

Magnetic field of a circular current loop

- The magnetic field along the axis of a loop of radius a carrying a current I is given by the equation below.
- The direction is given by the right-hand rule shown.

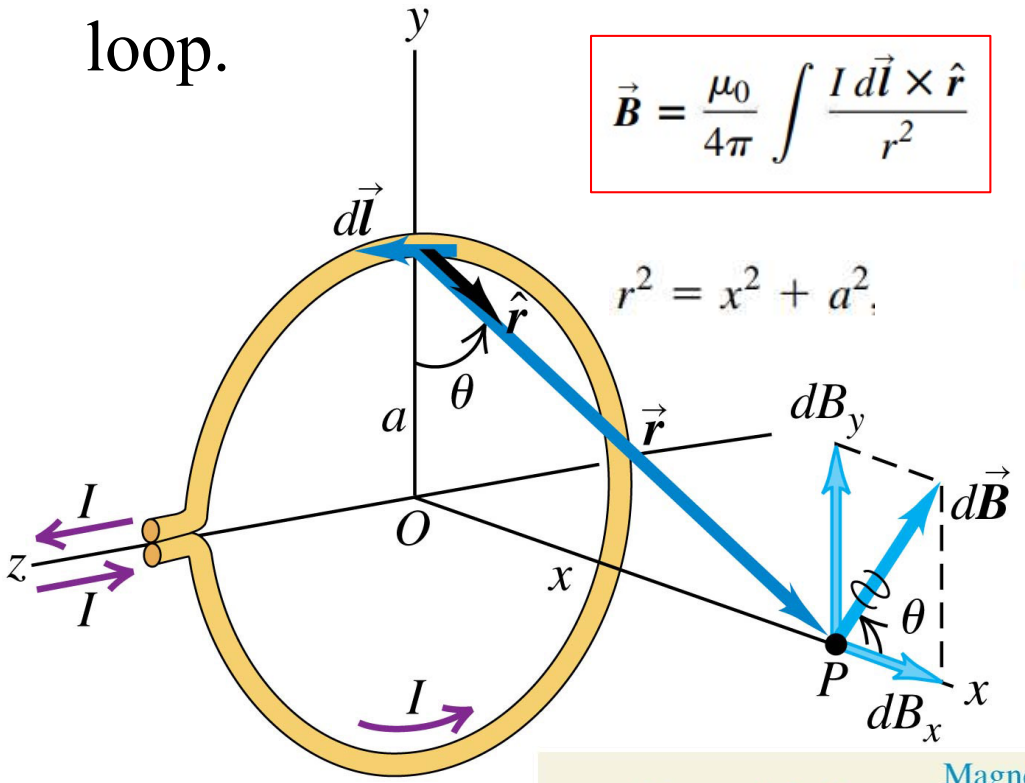




© 2016 Pearson Education, Ltd.

Magnetic field of a circular current loop

- Shown is a circular conductor with radius a carrying a counterclockwise current I.
- We wish to calculate the magnetic field on the axis of the loop.



$$\vec{B} = \frac{\mu_0}{4\pi}\int \frac{I\,d\vec{l}\times\hat{r}}{r^2}$$

$$dB = \frac{\mu_0 I}{4\pi}\frac{dl}{(x^2+a^2)}$$

$$r^2 = x^2 + a^2.$$

$$dB_x = dB\cos\theta = \frac{\mu_0 I}{4\pi}\frac{dl}{(x^2+a^2)}\frac{a}{(x^2+a^2)^{1/2}}$$

$$B_x = \int \frac{\mu_0 I}{4\pi}\frac{a\,dl}{(x^2+a^2)^{3/2}}$$

$$= \frac{\mu_0 Ia}{4\pi(x^2+a^2)^{3/2}}\int dl$$

Magnetic field on axis of a circular current-carrying loop

$$B_x = \frac{\mu_0 Ia^2}{2(x^2+a^2)^{3/2}}$$

Magnetic constant; Current; Radius of loop; Distance along axis from center of loop to field point

© 2016 Pearson Education, Ltd.

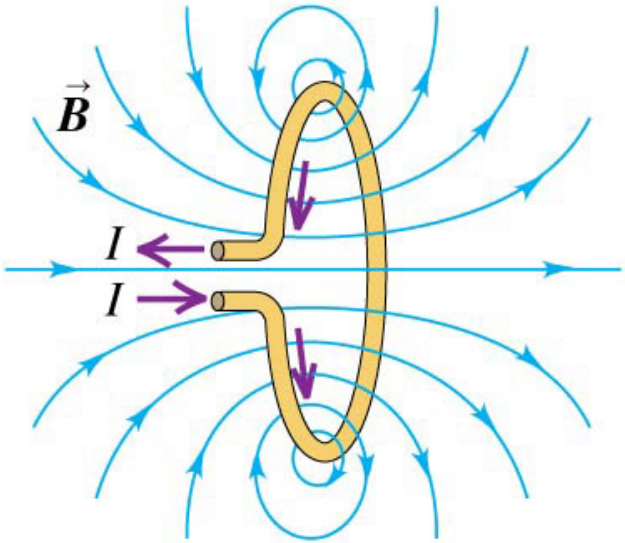
Question 13

- Find the position of zero field



© 2016 Pearson Education, Ltd.

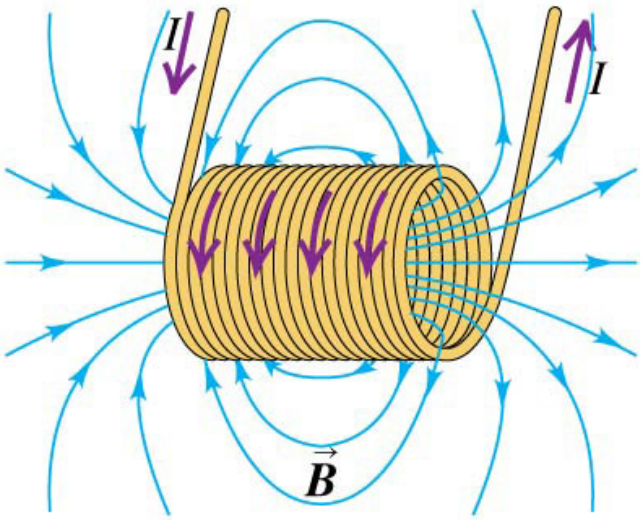
Magnetic field of circular current loops



$$B_x = \frac{\mu_0 I a^2}{2(x^2 + a^2)^{3/2}}$$

(circular loop)

$x = 0$ $\quad B_x = \dfrac{\mu_0 I}{2a}$



Solenoid

$$B_x = \frac{\mu_0 N I}{2a}$$

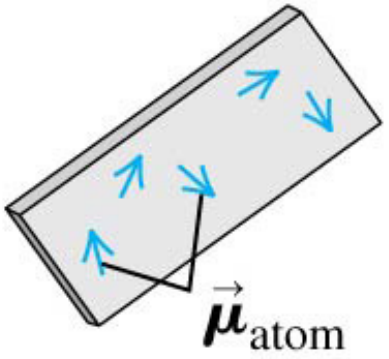
(center of N circular loops)

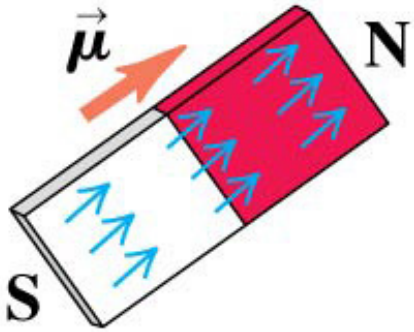
© 2016 Pearson Education, Ltd.

How magnets work

- **(a) An unmagnetized piece of iron**. Only a few representative atomic moments are shown.

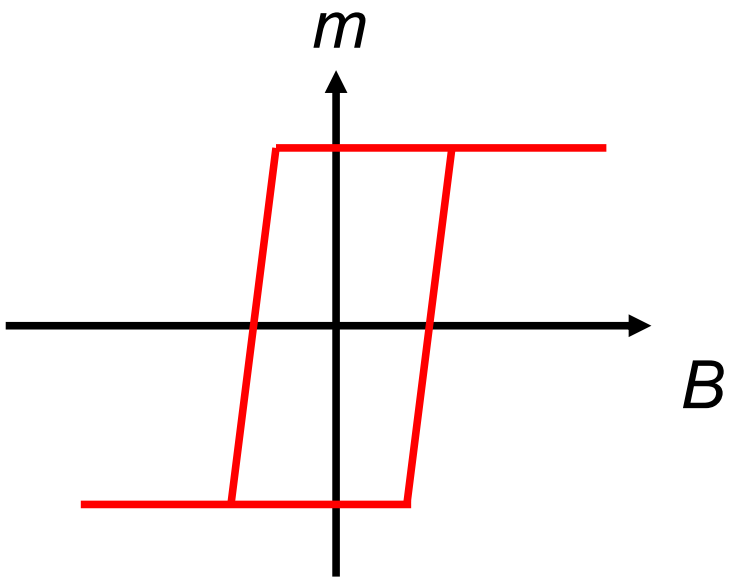
$$\vec{\mu} = \sum \vec{\mu}_{\text{atom}} = 0$$





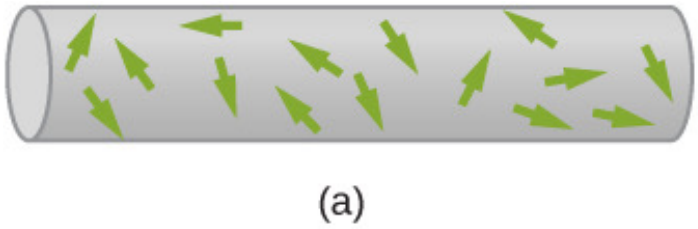
- **(b) A magnetized piece of iron** (bar magnet). The net magnetic moment of the bar magnet points from its south pole to its north pole.

$$\vec{\mu} = \sum \vec{\mu}_{\text{atom}} \neq 0$$

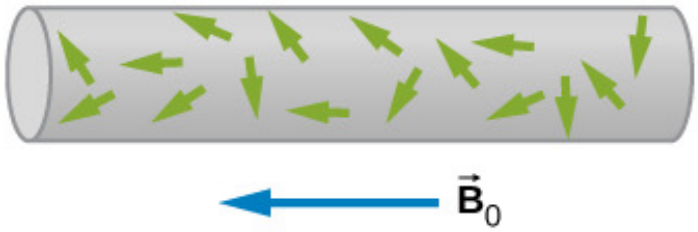


© 2016 Pearson Education, Ltd.

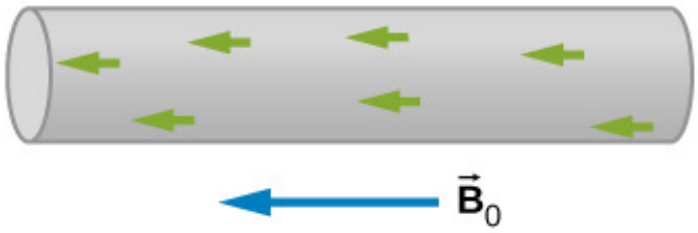
Dipoles in a magnet



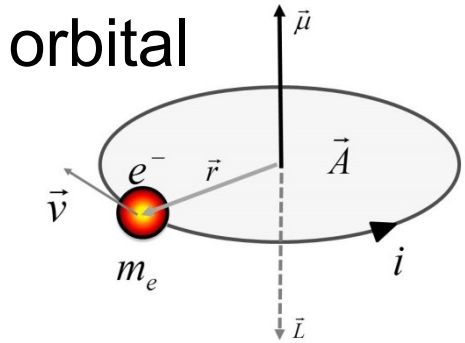
(a)

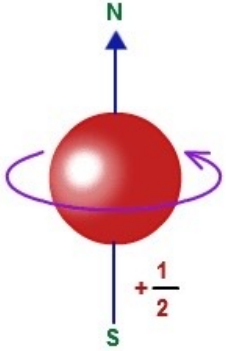


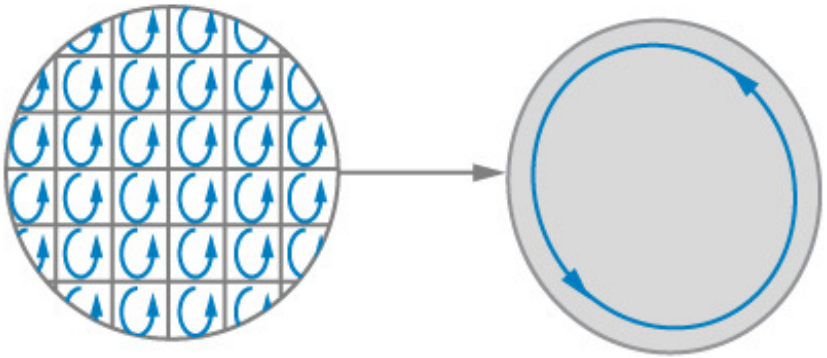
(b)

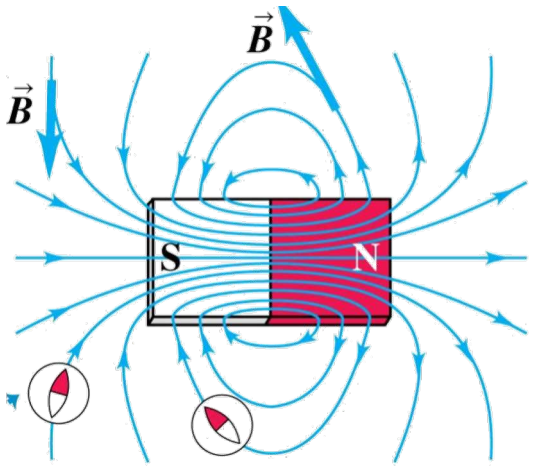


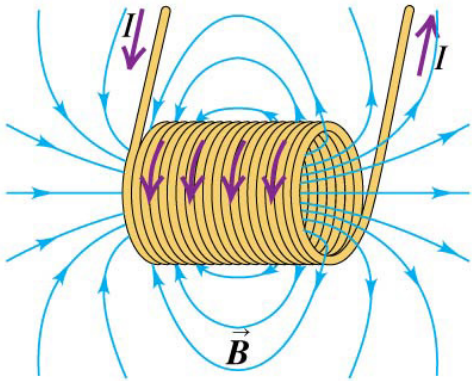
(c)









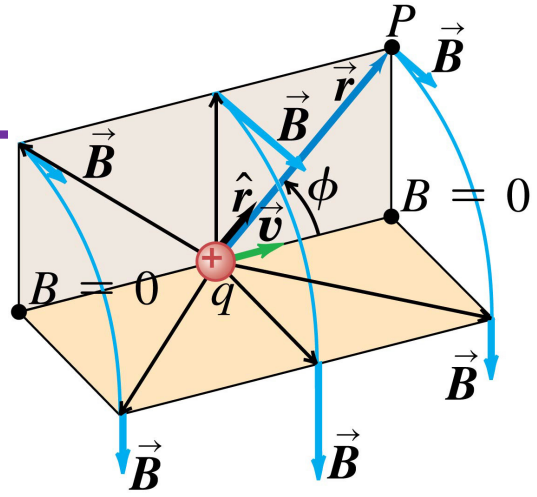


© 2016 Pearson Education, Ltd.

Summary

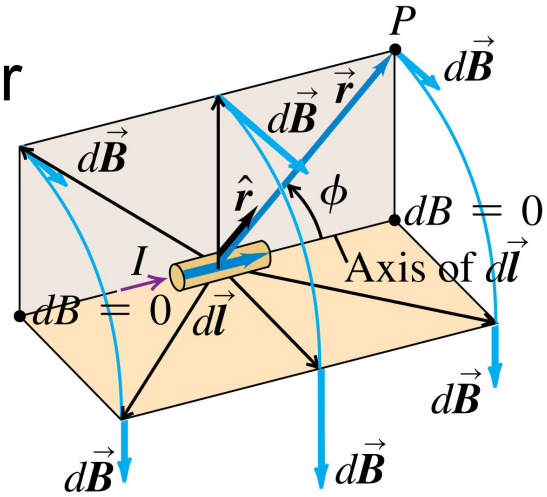
Magnetic field of a moving charge

$$\vec{B} = \frac{\mu_0}{4\pi}\frac{q\vec{v}\times\hat{r}}{r^2}$$



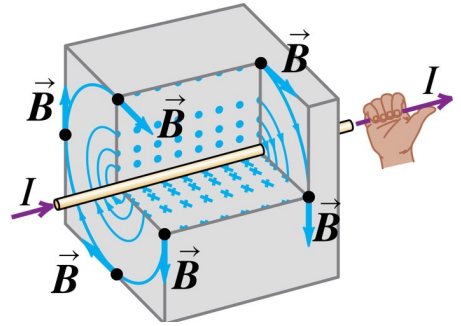
Magnetic field of a current-carrying conductor

$$d\vec{B} = \frac{\mu_0}{4\pi}\frac{I\,d\vec{l}\times\hat{r}}{r^2}$$



Magnetic field of a long, straight, current-carrying conductor

$$B = \frac{\mu_0 I}{2\pi r}$$

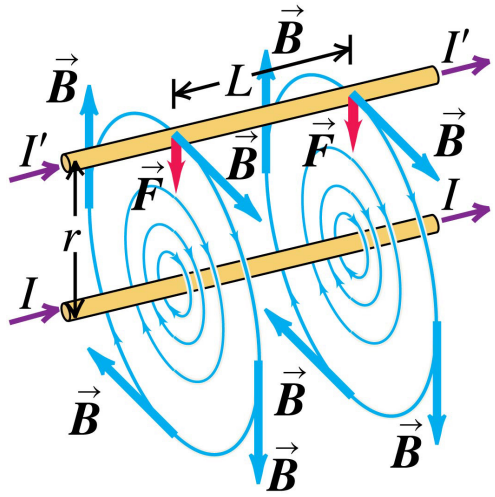


© 2016 Pearson Education, Ltd.

Summary

Magnetic force between current-carrying conductors

$$\frac{F}{L} = \frac{\mu_0 I I'}{2\pi r}$$



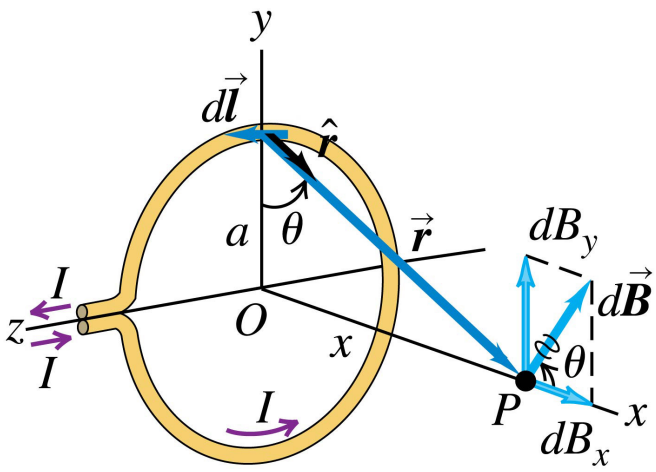
Magnetic field of a current loop

$$B_x = \frac{\mu_0 I a^2}{2(x^2 + a^2)^{3/2}}$$

(circular loop)

$$B_x = \frac{\mu_0 N I}{2a}$$

(center of N circular loops)



© 2016 Pearson Education, Ltd.

[참고]Old definition of the ampere (before 2019)

- The SI definition of the **ampere** is:

  One ampere is that unvarying current that, if present in each of two parallel conductors of infinite length and one meter apart in empty space, causes each conductor to experience a force of exactly 2×10^{-7} newtons per meter of length.

- This definition of the ampere is what leads us to choose the value of $4\pi \times 10^{-7}\ \mathrm{T \cdot m/A}$ for the magnetic constant, μ_0.

- The SI definition of the **coulomb** is the amount of charge transferred in one second by a current of one ampere.

Magnetic force per unit length between two long, parallel, current-carrying conductors

$$\frac{F}{L} = \frac{\mu_0 I I'}{2\pi r}$$

- μ_0: Magnetic constant
- I: Current in first conductor
- I': Current in second conductor
- r: Distance between conductors

© 2016 Pearson Education, Ltd.

[참고]New definition of the ampere (after 2019)

- The SI definition of the **ampere** is:

- **One ampere** is defined by taking the fixed numerical value of the elementary charge ***e*** to be $1.602176634\times10^{-19}$ when expressed in the unit C, which is equal to A·s, where the second is defined in terms of $\Delta\nu_{Cs}$.

$$1\ \mathrm{A} = \frac{e\,\Delta\nu_{Cs}}{(1.602\ 176\ 634\times10^{-19})(9\ 192\ 631\ 770)}$$

$$1\ \mathrm{C} = \frac{e}{1.602\ 176\ 634\times10^{-19}}$$

© 2016 Pearson Education, Ltd.